

The Behavior of Interest Rates

Chapter 5

Econ 351 – Monetary Economics

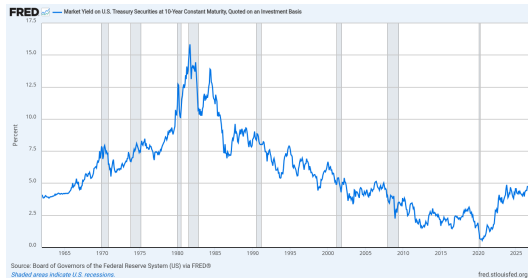
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Preview

This chapter explains how nominal interest rates are determined and why they change.

Tool: supply and demand in the **bond market**. Shifts in bond demand (wealth, expected returns, risk, liquidity) and bond supply (profitability, expected inflation, government deficits) move the interest rate.



The 10-year U.S. Treasury yield since 1962: interest rates move a lot.

Source: FRED, Federal Reserve Bank of St. Louis

Outline

Determinants of Asset Demand

Supply and Demand in the Bond Market

Changes in Equilibrium Interest Rates

Competing Views on Interest Rate Determination

Classical View (Goods Market)

- ▶ Equilibrium condition: $I = S$

Keynesian View (Money Market)

- ▶ Equilibrium condition: $M^s = M^d$

Where we are going: the classical loanable-funds view becomes a *bond market*: firms that invest supply bonds, households that save demand bonds. The price of bonds pins down the interest rate.

Classical Theory of Interest Rate Determination

Equilibrium Condition

$$C + S = C + I \Rightarrow S = I$$

Supply of Loanable Funds

$$S = S(i)$$

- ▶ Saving is an increasing function of the interest rate.

Demand for Loanable Funds

$$I = I(i)$$

- ▶ Investment is a decreasing function of the interest rate.

From Loanable Funds to Bond Supply and Demand

Bond Supply (Investment)

- ▶ Firms that want to invest demand funds and therefore supply bonds.
- ▶ $I(i) \rightarrow B^s(i) \rightarrow B^s(P)$

Bond Demand (Saving)

- ▶ Households that save supply funds and therefore demand bonds.
- ▶ $S(i) \rightarrow B^d(i) \rightarrow B^d(P)$

Bond price and interest rate move in opposite directions: a higher bond price P means a lower interest rate i .

Determinants of Asset Demand

What Is an Asset?

- ▶ People hold many different types of assets.
- ▶ An **asset** is anything that can be owned and has value.
- ▶ Examples: money, bonds, stocks, land, houses, and machinery.

Determinants of Asset Demand

- ▶ **Price factor** → movement along the demand curve
 - ▶ Expected return
- ▶ **Other factors** → shifts in the demand curve
 - ▶ Wealth
 - ▶ Risk
 - ▶ Liquidity

Theory of Portfolio Choice

Holding all other factors constant:

Variable	Change in Variable	Change in Quantity Demanded
Wealth	↑	↑
Expected return relative to other assets	↑	↑
Risk relative to other assets	↑	↓
Liquidity relative to other assets	↑	↑

Response of the quantity of an asset demanded to changes in wealth, expected returns, risk, and liquidity.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Outline

Determinants of Asset Demand

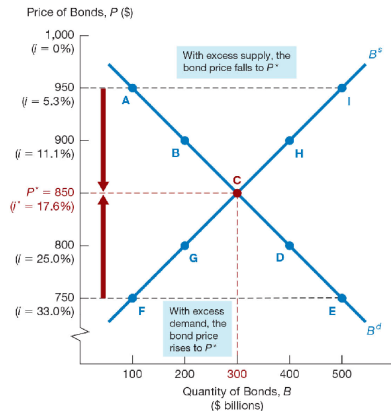
Supply and Demand in the Bond Market

Changes in Equilibrium Interest Rates

Supply and Demand in the Bond Market

Market Equilibrium

- ▶ Equilibrium occurs when bond demand equals bond supply.
- ▶ $B^d = B^s$ determines the equilibrium bond price and interest rate.
- ▶ If $B^d > B^s$: excess demand $\rightarrow$ bond price rises, interest rate falls.
- ▶ If $B^d < B^s$: excess supply $\rightarrow$ bond price falls, interest rate rises.



Supply and demand for bonds.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Outline

Determinants of Asset Demand

Supply and Demand in the Bond Market

Changes in Equilibrium Interest Rates

Shifts in the Bond Demand Curve: Wealth

Wealth and Bond Demand

- ▶ The main wealth-related factor affecting bond demand is saving.
- ▶ Higher saving increases the demand for bonds.

How Saving Increases

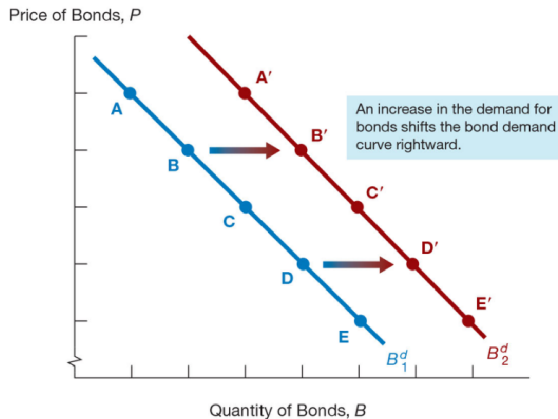
- ▶ Higher income
- ▶ Greater preference for saving

Increase in wealth $\rightarrow$ higher saving $\rightarrow$ greater bond demand $\rightarrow$ demand curve shifts right.

Shifts in the Bond Demand Curve: Other Factors

- ▶ **Expected returns:** Higher expected future interest rates lower expected returns on long-term bonds → demand shifts left.
- ▶ **Expected inflation:** Higher expected inflation lowers expected returns on bonds → demand shifts left.
- ▶ **Risk:** Higher riskiness of bonds → demand shifts left.
- ▶ **Liquidity:** Greater liquidity of bonds → demand shifts right.

Shifts in the Bond Demand Curve

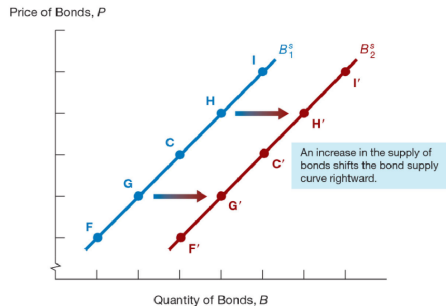


Shift in the demand curve for bonds.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Shifts in the Bond Supply Curve

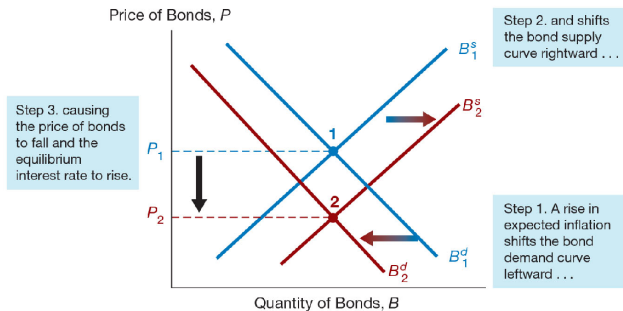
- ▶ **Expected profitability of investment opportunities:** In an expansion, the supply curve shifts to the right.
- ▶ **Expected inflation:** An increase in expected inflation shifts the supply curve for bonds to the right.
- ▶ **Government budget:** Increased budget deficits shift the supply curve to the right.



Shifts in the bond supply curve.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

The Fisher Effect: A Rise in Expected Inflation



Response to a change in expected inflation: demand shifts left, supply shifts right, the interest rate rises.

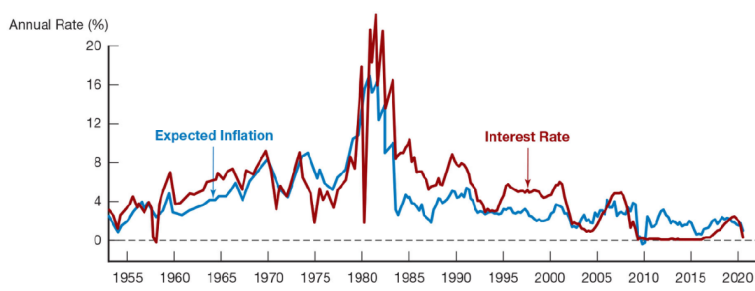
Source: Mishkin, *The Economics of Money, Banking and Financial Markets*



Irving Fisher (1867–1947), the economist behind the Fisher effect.

Wikimedia Commons, George Grantham Bain Collection, Public domain

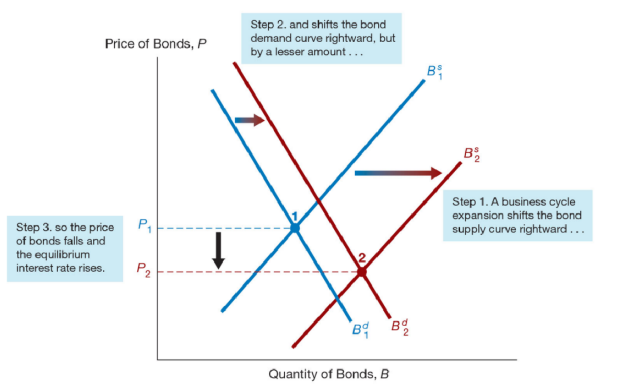
Expected Inflation and Interest Rates, 1953–2020



Expected inflation and the three-month Treasury bill rate move together, as the Fisher effect predicts.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

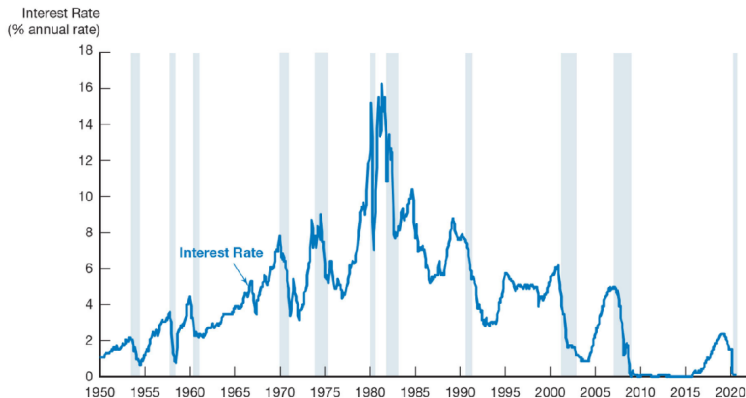
Response to a Business Cycle Expansion



In an expansion both supply and demand shift right; the interest rate usually rises.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Business Cycle and Interest Rates, 1951–2020



Three-month Treasury bill rate; shaded areas are recessions. Rates tend to rise in expansions and fall in recessions.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Chapter summary

- ▶ Asset demand depends on wealth, expected return, risk, and liquidity.
- ▶ The interest rate is determined by supply and demand in the bond market; bond prices and interest rates move in opposite directions.
- ▶ Bond demand shifts with wealth, expected returns, expected inflation, risk, and liquidity; bond supply shifts with expected profitability, expected inflation, and government deficits.
- ▶ Fisher effect: higher expected inflation raises the nominal interest rate.
- ▶ In a business cycle expansion both curves shift right and interest rates usually rise.